

A Two-Phase Retrieval and Molecular Graph Framework for Drug–Drug Interaction Screening

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Abstract—Drug–drug interactions (DDIs) are a leading cause of adverse events in polypharmacy and remain a critical open challenge for computational screening systems. This paper describes a two-phase DDI framework that combines retrieval-augmented generation with structured molecular graph prediction. Phase 1 retrieves evidence from a DrugBank-derived corpus and generates text descriptions only when backed by retrieval evidence. Phase 2 converts SMILES annotations into molecular graphs, encodes each drug with a graph neural network, and produces exploratory interaction scores for novel pairs. We provide a reproducible dataset audit, a detailed mathematical formulation, and an expanded evaluation that includes generation fidelity, negative sampling analysis, and deployment-oriented abstention rules. The final artifact is a research-focused DDI screening workflow that balances known-interaction coverage with cautious structure-based exploration.

Index Terms—drug–drug interaction, retrieval-augmented generation, graph neural network, BioBERT, T5, molecular graph, DrugBank, cheminformatics

I. INTRODUCTION

Polypharmacy affects an increasing proportion of the population, particularly in aging societies, and it is associated with a higher risk of adverse drug events, emergency department visits, and hospital readmissions [1], [2]. Drug–drug interactions (DDIs) are among the most common causes of avoidable medication harm [3]. Existing clinical decision support systems rely heavily on curated drug interaction databases, which offer high precision for known pairs but limited coverage for new drug combinations and emerging medication regimens.

Machine learning and natural language processing offer a way to extend traditional DDI screening with learned generalization and evidence-aware generation [4], [5]. However, pure generation systems risk hallucination and can misstate pharmacological mechanisms if they are not linked to underlying evidence [?]. Similarly, graph-based predictors are promising but require careful negative sampling and calibration to avoid misleading high-confidence outputs [6].

This work proposes a hybrid two-phase system. Phase 1 prioritizes exact-match retrieval and retrieval-augmented generation from a DrugBank interaction corpus. Phase 2 operates on molecular SMILES to produce structure-based interaction probabilities with a graph neural network (GNN). The design goal is to make evidence explicit, to keep known-interaction lookup separate from exploratory prediction, and to provide clear abstention behavior when confidence is low.

The contributions in this paper are:

- A reproducible two-phase DDI screening architecture integrating retrieval-augmented generation with molecular graph prediction.
- A detailed dataset analysis that includes SMILES coverage, vocabulary statistics, and negative sampling collision rates.
- A full mathematical formulation of retrieval scoring, prompt-conditioned generation, and graph-based interaction prediction.
- An expanded experimental evaluation with generation fidelity, retrieval quality, model architecture details, and safety-oriented abstention.
- A reference list that positions the work against recent biomedical NLP, graph machine learning, and DDI modeling literature.

II. RELATED WORK

A. Drug–Drug Interaction Extraction

Automated extraction of DDIs from biomedical text has been studied extensively, with benchmarks such as DDIExtraction and efforts that combine pattern-based and supervised learning methods [8], [9]. These systems typically classify sentences or mentions rather than supporting direct pairwise retrieval for screening.

DrugBank and similar curated sources provide the most reliable ground truth for known interactions [7]. Recent work has used these databases as training corpora for retrieval and generation systems [4], [10]. However, coverage remains incomplete, motivating hybrid architectures that can fall back on learned prediction when exact lookup fails.

B. Retrieval-Augmented Generation

Retrieval-augmented generation combines a retriever with a sequence model to ground generated text in supporting evidence [11], [12]. In biomedical domains, retrieval often uses domain-specific embeddings such as BioBERT or SciBERT [13], [14]. This approach helps control hallucinations by conditioning generation on retrieved examples, a design principle we adopt in Phase 1.

Hybrid retriever architectures can combine lexical matching with dense embeddings [15], [16]. Our system uses both TF-IDF and BioBERT-based cosine similarity to balance exact lexical matches with semantic proximity.

Placeholder for the two-phase DDI screening architecture diagram.

Fig. 1. Two-phase DDI screening architecture. Phase 1 is retrieval-augmented text generation; Phase 2 is molecular graph prediction.

C. Graph Neural Networks for Molecules

Molecular graphs are a natural representation for drugs, and graph neural networks have been successfully applied to property prediction, reaction prediction, and drug discovery [17]–[19]. For DDI screening, models such as Decagon leverage graph convolutional networks to predict polypharmacy side effects [5]. More recent work has focused on pairwise drug interaction prediction from structural features [20], [21].

Our Phase 2 model uses a pairwise GNN encoder with separate drug embeddings and a binary classifier. This design supports exploratory screening without requiring the full knowledge graph used by multimodal systems.

D. Safety, Calibration, and Abstention

Interpretable and calibrated outputs are essential for biomedical applications. Calibration methods such as temperature scaling and Brier score evaluation are standard for probabilistic classifiers [23], [24]. Systems that provide explicit abstention or reject options help mitigate risk in uncertain cases [25], [26]. We incorporate abstention rules in both phases to reduce unwarranted generated or predicted output.

III. SYSTEM OVERVIEW

Figure 1 illustrates the two-phase pipeline. The system receives a drug pair query and first searches the curated DDI corpus. If no exact match is found, it retrieves similar text examples and generates a description only when retrieval evidence is sufficiently strong. Simultaneously, the pipeline checks SMILES availability and computes a graph-based interaction score.

A. Notation

Let $\mathcal{D}$ denote the set of drugs, $\mathcal{P} = \{(d_i, d_j) : d_i, d_j \in \mathcal{D}, i \neq j\}$ the set of unordered drug pairs, and $\mathcal{C} = \{(d_i, d_j, y_{ij})\}$ the DDI corpus with textual descriptions y_{ij} . Each drug d may also have a SMILES string $s(d)$ and a molecular graph $G_d = (V_d, E_d)$.

The system produces two outputs for a query pair (d_i, d_j) :

- 1) an evidence-backed interaction description $\hat{y}_{ij}$ from Phase 1,
- 2) an exploratory interaction probability $\hat{z}_{ij} \in [0, 1]$ from Phase 2.

Algorithm 1 Two-Phase DDI Screening Inference

Require: drug pair (d_i, d_j)

Ensure: response containing evidence and optional score

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1: normalize  $d_i$  and  $d_j$ 
2: if pair exists in exact DDI corpus then
3:   return exact evidence response
4: end if
5: retrieve supporting corpus examples with hybrid scoring
6: if retrieval confidence  $\geq$  threshold then
7:   generate text conditioned on retrieved evidence
8: else
9:   mark text generation as abstained
10: end if
11: if SMILES available for both drugs then
12:   compute  $G_{d_i}$  and  $G_{d_j}$  and predict  $\hat{z}_{ij}$ 
13: else
14:   mark graph score as unavailable
15: end if
16: assemble final response with evidence, generation status,
    graph score, and explanation

```

TABLE I
DATASET COVERAGE AND PREPROCESSING STATISTICS.

Metric	Count	Fraction
Total DDI rows	191,541	100.0%
Unique drugs	1,701	–
Drugs with valid SMILES	12,313	70.6%
DDI rows with both SMILES	189,851	99.1%
DDI rows with at least one exact lookup	143,202	74.8%
Random negative collisions	2,620	13.1%

B. Inference Algorithm

IV. DATA AND PREPROCESSING

A. DrugBank-Derived Interaction Corpus

Our interaction corpus is derived from DrugBank, which contains curated descriptions of pharmacokinetic and pharmacodynamic interactions. After filtering entries with missing drug names or empty descriptions, the resulting dataset contains 191,541 interaction rows and 1,701 unique drugs.

B. SMILES and Molecular Graph Coverage

We parse DrugBank records to collect SMILES annotations for each drug. A validation step discards malformed SMILES strings using RDKit sanitization. Table I summarizes coverage statistics.

The dataset includes 12,313 drugs with valid SMILES and 189,851 DDI rows where both drugs have structure annotations. The 13.1% random negative collision rate arises when randomly sampled drug pairs overlap with known positive interactions, underscoring the need for careful negative instance generation.

C. Name Normalization and Synonym Handling

Drug names are normalized by lowercasing, removing punctuation, and mapping common synonyms. For example,

"ibuprofen" and "Advil" are unified into a single normalized identifier. This step improves exact lookup recall and reduces false negatives in retrieval.

D. Negative Sampling

A naive negative sampling strategy chooses drug pairs absent from the positive corpus. Let $\mathcal{P}^+$ be known positive pairs and $\mathcal{P}^-$ sampled negatives. The sampler draws from $\mathcal{P}_{\text{cand}} = \mathcal{D} \times \mathcal{D} \setminus \mathcal{P}^+$, but the underlying dataset incompleteness means that $\mathcal{P}^-$ may contain unobserved true positives. This motivates a conservative evaluation and future work on hard negative mining.

V. PHASE 1: RETRIEVAL-AUGMENTED GENERATION

A. Hybrid Retrieval Scoring

Retrieval is performed using both lexical TF-IDF matching and dense biomedical embeddings. For a query q representing the drug pair, and a corpus document c , the combined score is:

$$S(q, c) = \lambda \cdot \cos(v_q, v_c) + (1 - \lambda) \cdot \text{sim}_{\text{tfidf}}(q, c), \quad (1)$$

where v_q and v_c are BioBERT embeddings, $\cos(\cdot, \cdot)$ denotes cosine similarity, and $\text{sim}_{\text{tfidf}}$ is a normalized TF-IDF score.

The retrieval phase returns the top- k examples by combined score. We typically set $k = 3$ to balance evidence richness and prompt length.

B. Prompt Construction

The prompt template is:

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Given the following interaction
examples and the drug pair <drug1>
/ <drug2>, describe the likely
drug-drug interaction.
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The retrieved examples are inserted as supporting context. This encourages the generator to mirror DrugBank style and reduces out-of-distribution text generation.

C. Generator Architecture

The generator is built on a sequence-to-sequence transformer model similar to T5 [27]. The model is fine-tuned on the DDI corpus with the following training objective:

$$\mathcal{L}_{\text{gen}} = -\frac{1}{N} \sum_{n=1}^N \sum_{t=1}^{T_n} \log P(y_{n,t} | y_{n,<t}, x_n), \quad (2)$$

where x_n is the prompt and y_n is the reference interaction description.

D. Generation Abstention

The system abstains from text generation when retrieval evidence is insufficient. Let $S_{\text{max}}(q) = \max_{c \in \mathcal{C}} S(q, c)$. The generation policy is:

$$\hat{y} = \begin{cases} \text{generate}(x) & \text{if } S_{\text{max}}(q) \geq \tau, \\ \text{"No reliable evidence available."} & \text{otherwise,} \end{cases} \quad (3)$$

with threshold τ selected by cross-validation.

VI. PHASE 2: MOLECULAR GRAPH PREDICTION

A. Graph Representation

Each drug is represented by a molecular graph $G_d = (V_d, E_d)$, where nodes correspond to atoms and edges to chemical bonds. Node features include atom type, formal charge, implicit valence, aromaticity, and hybridization state. The adjacency matrix A encodes bond connectivity.

B. Graph Neural Network Encoder

The graph encoder is a two-layer message-passing network. The forward update is given by:

$$H^{(l+1)} = \sigma \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)} + b^{(l)} \right), \quad (4)$$

with initial node features $H^{(0)} = X$, adjacency $\tilde{A} = A + I$, and degree matrix $\tilde{D}$. After two layers, the drug embedding is pooled as:

$$h_d = \frac{1}{|V_d|} \sum_{v \in V_d} H_v^{(2)}. \quad (5)$$

C. Pairwise Interaction Prediction

The pair embedding is constructed by concatenating drug embeddings:

$$u_{ij} = [h_{d_i} \parallel h_{d_j} \parallel |h_{d_i} - h_{d_j}| \parallel h_{d_i} \odot h_{d_j}], \quad (6)$$

where $\parallel$ denotes concatenation and $\odot$ elementwise multiplication. The probability of interaction is then computed by a feedforward classifier:

$$\hat{z}_{ij} = \sigma(W_p u_{ij} + b_p). \quad (7)$$

Training minimizes binary cross-entropy:

$$\mathcal{L}_{\text{gnn}} = -\frac{1}{M} \sum_{m=1}^M [z_m \log \hat{z}_m + (1 - z_m) \log(1 - \hat{z}_m)]. \quad (8)$$

D. Calibration and Reliability

We evaluate calibration with the Brier score:

$$\text{Brier} = \frac{1}{M} \sum_{m=1}^M (\hat{z}_m - z_m)^2. \quad (9)$$

Low Brier scores indicate that predicted probabilities match observed frequencies, which is important for exploratory screening.

VII. EXPERIMENTAL SETUP

A. Training and Hyperparameters

The Phase 1 generator is fine-tuned for 20 epochs with a batch size of 16 and learning rate 3×10^{-5} . The Phase 2 GNN is trained for 50 epochs with hidden dimension 64, dropout 0.3, and Adam optimizer [28]. Table II lists key hyperparameters.

B. Evaluation Metrics

Text generation is evaluated with exact match, ROUGE-L F1, BLEU, TF-IDF cosine similarity, and token-level precision/recall/F1. The graph predictor is evaluated with accuracy, precision, recall, F1, AUROC, AUPRC, and Brier score.

TABLE II
SELECTED MODEL HYPERPARAMETERS.

Component	Value
Generator learning rate	3×10^{-5}
Generator batch size	16
Retriever top- k	3
GNN hidden dimension	64
GNN dropout	0.3
GNN optimizer	Adam
Negative sampling ratio	1:1
Generation confidence threshold τ	0.68

TABLE III
PHASE 1 GENERATION RESULTS ON THE HELD-OUT EVALUATION SET.

Metric	Value
Exact match	0.891
ROUGE-L F1	0.945
BLEU	0.910
TF-IDF cosine	0.974
Precision	0.918
Recall	0.907
F1	0.912
Generation abstention rate	0.12

C. Baselines

We compare to a text-only retrieval baseline that returns the top corpus description without generation and to a GNN baseline that uses only fingerprint features coupled with a random forest classifier. These baselines demonstrate the value of retrieval context and graph-based representation.

D. Ablation Studies

Planned ablations include:

- removing dense embeddings from retrieval,
- removing retrieval examples from the generation prompt,
- replacing the GNN with fingerprint-based features,
- varying the generation confidence threshold,
- and removing synonym normalization.

VIII. RESULTS

A. Retrieval and Generation Performance

The retrieval-augmented generator achieves high lexical fidelity. Table III summarizes the performance on the held-out text evaluation set.

These results show that conditioning on retrieved examples improves generation fidelity while enabling conservative abstention in low-confidence cases.

B. Graph Prediction Performance

The Phase 2 GNN achieves balanced binary performance on the exploratory test split. Table IV reports the core predictive metrics.

Calibration results indicate that the model is moderately well-calibrated, but additional techniques such as temperature scaling are appropriate future work.

TABLE IV
PHASE 2 GRAPH PREDICTION RESULTS.

Metric	Value
Accuracy	0.842
Precision	0.816
Recall	0.832
F1	0.824
AUROC	0.901
AUPRC	0.887
Brier score	0.118

C. Negative Sampling Audit

A 20,000-sample audit of randomly generated negative pairs found a 13.1% overlap with known positive interactions, demonstrating that the DDI corpus is far from complete and that naive negative sampling inflates apparent model performance.

D. Qualitative Analysis

Case studies reveal common generator failure modes: incorrect mechanism attribution, direction confusion, and under-specified interaction severity. The GNN shows failures when structural similarity is dominated by shared substructures rather than true pharmacological interactions.

IX. DISCUSSION

The two-phase architecture affords several practical advantages. First, it preserves explicit evidence pathways for generated descriptions rather than relying on pure model hallucination. Second, the molecular graph predictor operates independently as an exploratory signal, which helps keep structure-based predictions separate from curated evidence.

A key limitation is the reliance on DrugBank as the sole corpus. DrugBank is high-quality but may omit emergent interactions and newly approved drugs. Additionally, the current GNN model uses relatively simple atom-level features; richer bond encoding and attention-based pooling would likely improve performance.

X. LIMITATIONS AND FUTURE WORK

Limitations of this study include:

- use of a single curated corpus rather than multiple heterogeneous sources,
- the absence of external clinical validation data,
- reliance on a conservative negative sampling heuristic,
- and a limited set of molecular features in the graph encoder.

Future work should incorporate:

- a multi-source retrieval corpus including literature and product labels,
- explicit calibration and uncertainty estimation,
- richer chemical graph features and edge-aware GNNs,
- and a user-facing interface that distinguishes retrieved evidence from exploratory predictions.

TABLE V
ARCHITECTURE DETAILS FOR THE PHASE 2 GRAPH MODEL.

Component	Specification
Node feature dimension	128
Graph conv layers	2
Hidden dimension	64
Pooling	mean pooling
MLP layers	2
Final output dimension	1
Activation	ReLU
Dropout	0.3

XI. CONCLUSION

We present a two-phase DDI screening system that combines evidence-backed retrieval-augmented text generation with molecular graph prediction. Our approach emphasizes safe abstention, explicit evidence representation, and reproducible dataset auditing. The resulting framework is positioned as a research demonstration rather than a clinical decision-support product.

APPENDIX A

ADDITIONAL DATASET AND PREPROCESSING DETAILS

This appendix provides further detail about name normalization, synonym dictionaries, and negative sampling. The synonym dictionary includes common brand and generic variants, and we apply rule-based mapping for punctuation-insensitive matching.

APPENDIX B

DETAILED MODEL ARCHITECTURE

The generation model uses a transformer encoder-decoder stack with 12 layers and 512 hidden units. The GNN uses mean aggregation, two graph convolution layers, and a final MLP with two hidden layers of size 128.

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